STAT 639 (Spring 2026): Homework 4

Due: April 21, 11:59pm Central Time, via Canvas

ISLR stands for the book "An Introduction to Statistical Learning with Applications in R, Second Edition".

1. Ex.9 of Ch.8 in ISLR
2. Ex.10(a)–(e) of Ch.8 in ISLR
3. Ex.1 of Ch.10 in ISLR
4. Ex.4(a)–(b) of Ch.10 in ISLR (Assume that after the first convoluational layer, the three-dimensional feature maps are then flattened and then fed into the output layer consisting of a single node (for a regression problem).)
5. Prove the equality
$$\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}(x_{ij}-x_{i'j})^2 = 2\sum_{i\in C_k}\sum_{j=1}^{p}(x_{ij}-\bar{x}_{kj})^2$$
shown on page 13 of slide set 7.
6. Ex.3 of Ch.12 in ISLR